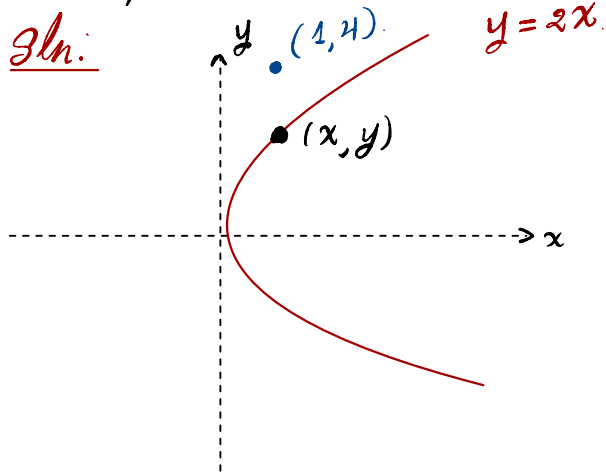


Example. Find the point on the parabola $y^2 = 2x$ that is closest to the point $(1, 4)$.



Using the formula for the distance between two points $(1, 4)$ and (x, y) .

Recall: (x_1, y_1) and (x_2, y_2)

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

then, using (x, y) and $(1, 4)$.

$$d = \sqrt{(x-1)^2 + (y-4)^2}$$

Restriction: $y^2 = 2x$. $y = \sqrt{2x}$, $x = \frac{y^2}{2}$.

So,

$$d(x) = \sqrt{(x-1)^2 + (\sqrt{2x} - 4)^2}$$

$$d(y) = \sqrt{\left(\frac{y^2}{2} - 1\right)^2 + (y-4)^2}$$

$$\left(((2x)^{1/2} - 4)^2 \neq ((2x)^{1/2})^2 - 4^2 \right)$$

$$(x-y)^2 \neq x^2 - y^2$$

Instead of minimizing d , we minimize its square.

$$d^2 = f(y) = \left(\frac{y^2}{2} - 1\right)^2 + (y-4)^2$$

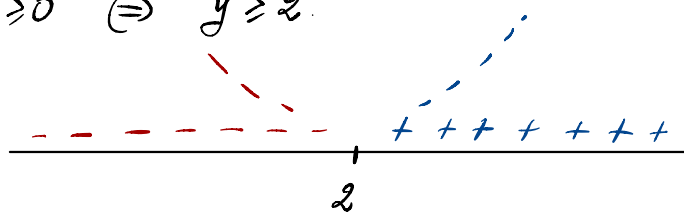
$$f'(y) = 2\left(\frac{y^2}{2} - 1\right)\left(\frac{2y}{2}\right) + 2(y-4)(1)$$

$$f'(y) = (y^2 - 2)(y) + 2y - 8 = y^3 - 2y + 2y - 8 = y^3 - 8$$

then $f'(y) = 0 \Rightarrow y = 2$.

$$f'(y) \geq 0 \Rightarrow y^3 - 8 \geq 0 \quad \underbrace{\geq 0}_{(y-2)(y^2+2y+4) \geq 0}$$

then $y-2 \geq 0 \Rightarrow y \geq 2$.

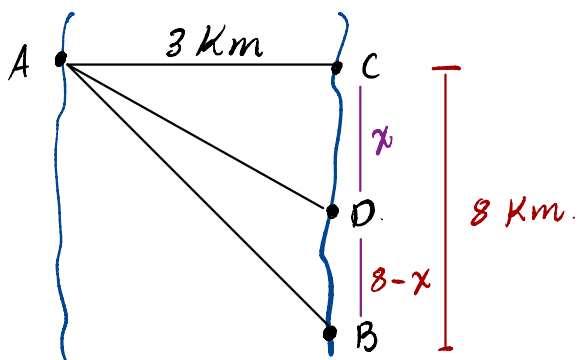


By the first derivative test at $y=2$ we have a local minimum.

The corresponding value of the x is $x = \frac{1}{2}y^2 = \frac{1}{2}(2)^2 = 2$. then the point on the parabola $y^2 = 2x$ closest to $(1,4)$ is $(2,2)$, the distance between the points is

$$\begin{aligned} d &= \sqrt{f(2)} = \sqrt{\left(\frac{2^2}{2} - 1\right)^2 + (2-4)^2} \\ &= \sqrt{\left(\frac{4}{2} - 1\right)^2 + (-2)^2} \\ &= \sqrt{(2-1)^2 + 4} \\ &= \sqrt{5} \end{aligned}$$

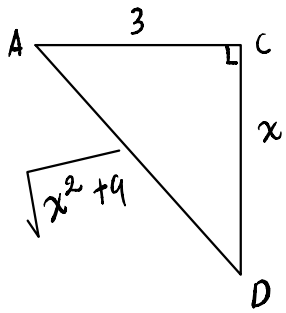
Example. A woman launches her boat from point A on a bank of straight river, 3km wide, and wants to reach point B, 8 km downstream on the opposite bank as quickly as possible. She could row her boat directly across the river to point C and then run to B, or she could row directly to B, or she could row to some point D between C and B and then run to B. If she can row 6 km/h and run 8 km/h where should she land to reach B as soon possible.



Sln:

$$\text{time} = \frac{\text{distance}}{\text{rate}}$$

$$|BD| = 8-x, \quad |AD| = \sqrt{x^2 + 9}$$



then, the rowing time is $\frac{\sqrt{x^2+9}}{6}$

and the running time is $\frac{(8-x)}{8}$ so, the function for the time is:

$$T(x) = \frac{\sqrt{x^2+9}}{6} + \frac{8-x}{8} ; \text{Dom } T = [0, 8]$$

$$\text{Then, } T'(x) = \frac{1}{6} \cdot \frac{d}{dx}(\sqrt{x^2+9}) + \frac{d}{dx}\left(\frac{8}{8} - \frac{x}{8}\right)$$

$$T'(x) = \frac{1}{6} \left\{ \frac{1}{2\sqrt{x^2+9}} \cdot \frac{d}{dx}(x^2+9) \right\} + 0 - \frac{d}{dx}\left(\frac{x}{8}\right)$$

$$T'(x) = \frac{1}{6} \left\{ \frac{\cancel{2}x}{\cancel{2}\sqrt{x^2+9}} \right\} - \frac{1}{8} = \frac{x}{6\sqrt{x^2+9}} - \frac{1}{8}$$

$$T'(x)=0 \Rightarrow \frac{x}{6\sqrt{x^2+9}} - \frac{1}{8} = 0$$

$$\Rightarrow \frac{x}{6\sqrt{x^2+9}} = \frac{1}{8}$$

$$\Rightarrow x = \frac{6}{8} \sqrt{x^2+9}$$

$$\Rightarrow x = \frac{3}{4} \sqrt{x^2+9}$$

$$\Rightarrow x^2 = \left(\frac{3}{4} \sqrt{x^2+9}\right)^2$$

$$\Rightarrow x^2 = \frac{9}{16} (x^2+9)$$

$$16x^2 = 9x^2 + 81$$

$$16x^2 - 9x^2 = 81$$

$$7x^2 = 81$$

$$x^2 = \frac{81}{7}$$

$$x = \pm \sqrt{\frac{81}{7}} = \pm \frac{9}{\sqrt{7}} \quad \text{then } x = \frac{9}{\sqrt{7}}$$

this is a critical point and by the theorem $x=0$ and $x=8$ are also critical points, we follow the closed interval method by evaluating T at all three points

$$T(0) = \frac{3}{6} + 1 = \frac{3}{2} = 1.5$$

$$T\left(\frac{9}{\sqrt{7}}\right) = 1 + \frac{\sqrt{7}}{8} \approx 1.33 \longrightarrow \text{The minimum.}$$

$$T(8) = \frac{\sqrt{3}}{6} \approx 1.42.$$

Since the smallest of these values of T occurs when $x = \frac{9}{\sqrt{7}}$ the absolute minimum value of T must occur there.

Then, the woman should land the boat at a point $\frac{9}{\sqrt{7}}$ km downstream from her starting point.

Problem for you! Find the area of the largest rectangle that can be inscribed in a semicircle of radius r .

